

Radio Resource Allocation Between Massive MIMO and LTE Using SDN

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Abstract—Massive multiple-input, multiple-output (Ma-MIMO) is currently being deployed in the fifth generation (5G) networks. It has increased the capacity for sub-6GHz wireless access. However, the ongoing development is a critical step in enhancing this technology in order to deliver the capacity gains and the Quality of Service (QoS) requirements in 5G. Here, a novel spatial Software-Defined Networking (SDN) controller framework for radio resource allocation between a single cell long-term evolution (LTE) and a single cell Ma-MIMO is proposed and shown to further enhance performance. For the first time, an SDN-based radio resource management (SDN-RRM) framework for multiple radio access technologies (multi-RATs) is introduced. The methodologies in this framework address the interference caused by the user channel vectors as well as hardware impairments in Ma-MIMO by mainly relying on the Error Vector Magnitude (EVM) and channel state information (CSI). The handover between the LTE Access Point (AP) and the Ma-MIMO AP is decided by the wireless SDN controller. This has led to maximize the throughput and to achieve the QoS requirement. The proposed framework has been built and evaluated in real-time by using a massive Ma-MIMO testbed and an LTE testbed controlled by an SDN controller.

Index Terms—Massive MIMO, 5G, User Grouping, Multi-RAT, SDN controller/agent, Quality of Service.

I. INTRODUCTION

Massive multiple-input, multiple-output (Ma-MIMO) is a Multi-user (MU) multiple-input, multiple-output (MIMO) architecture with a large number of antennas at the base station (BS) serving several users within the same time and frequency resource [1]. It has become a key fifth generation (5G) technology for sub-6 GHz wireless access due to its significant capacity gains. There is a large gap between the theoretical performance and what is achieved in practice. Industries are still improving this technology and delivering new record 5G capacity gains [2]. Radio resource management (RRM) plays a significant roll in boosting the capacity gains from Ma-MIMO, which is illustrated and experimentally demonstrated in [3]. This can be done by reallocating some users to different time slots or frequencies. Channel reciprocity is essential for downlink (DL) data transmission in Ma-MIMO time division duplex (TDD) systems. Not only propagation conditions determine the radio channel, but also differences in the analog transceiver

at both ends of the radio link. Since different transceiver chains are used at the BS and the user equipment (UE), there is a difference between the actual channel and the estimated channel [4].

Adapting the Software-Defined Networking (SDN) approach could help to increase the capacity gains from Ma-MIMO. The core idea of the SDN paradigm [5] is to separate the control plane from the data plane. By doing this, the controller makes the network forwarding decisions and directly controls the network to achieve desired operating behaviours, hence enhancing the network control programmability. Therefore, SDN is a natural enabler for developing innovative radio resource management schemes that decouple radio resource allocation decision functions from underlying wireless hardware in a simplified and improved way. According to the Open Networking Foundation (ONF) white paper [5], the SDN architecture consists of the application, control and data plane. The data plane is comprised of network elements with the Control-Data-Plane Interface (CDPI) agent to expose the capabilities of SDN datapaths to the controller.

In recent years, a significant amount of research work has been done to integrate SDN principles with wireless local area network (WLAN). Odin [6] is an SDN Wi-Fi network architecture based on the concept of light Virtual AP (LVAP), which separates states of each station association from physical APs. In [7] and [8], an SDN agent based framework is presented to extend the basic concept of SDN to WLANs. Using the SDN concept in Ma-MIMO has also started to be considered. The work in [9] has proposed spatial domain management for Ma-MIMO by using SDN as the central controller of the RRM in heterogeneous networks.

In this paper, we develop a framework that, to the best of our knowledge, is the first SDN-based radio resource management (SDN-RRM) framework for multiple radio access technologies (multi-RATs). The framework incorporates SDN agent with its Application Programming Interface (API)s to separate the control and data planes. The benefits of this SDN-RRM idea are three-fold: i) the framework breaks the boundaries among different wireless access technologies, hence providing a great

view of the aggregated resource of the underlying wireless network; ii) the framework offers a great degree of flexibility to dynamically satisfy different Quality of Service (QoS) requirements; and iii) building RRM algorithms in form of SDN control applications at the SDN controller side provides the ability of updating RRM functions on the fly as well as easier deployment and evolution.

A. Notation

Boldface (lower case) is used for column vectors, $\mathbf{x}$, and (upper case) for matrices, $\mathbf{X}$. The operators $(\cdot)^T$ and $(\cdot)^H$ denote transpose and Hermitian transpose. Let $\tilde{x}$ denotes the prediction of the next x value. The operator $\mathbb{E}\{\cdot\}$ denotes the expected value. The matrix $\mathbf{I}$ denotes the identity matrix. The set of the complex numbers and the set containing zero and the real positive numbers are denoted by $\mathbb{C}$ and $\mathbb{R}$, respectively. The Number and the set of active users are denoted by K and $\mathcal{U}$ respectively, where $\mathcal{U} = \{1, \dots, K\}$. Users who can only be served by a long-term evolution (LTE) Access Point (AP) or a Ma-MIMO AP are denoted by $(\cdot)^L$ and $(\cdot)^{Ma}$ respectively. Users overlapped between Ma-MIMO and LTE cells are denoted by $(\cdot)^{M-RAT}$, which can be served by either Ma-MIMO or LTE APs. While the overall users who can be served by Ma-MIMO AP are denoted by $(\cdot)^{MA}$. The majority estimation and prediction aspects of the paper are based on $(\cdot)^{MA}$. Thus, for notational simplicity, we consider $(\cdot) = (\cdot)^{MA}$.

II. SYSTEM MODEL

A single-cell Massive-MIMO and a single-cell LTE architecture are considered in this work, where both cells are overlapped as it is shown in Fig 1. Active users are served by either Ma-MIMO or LTE APs. All users can support LTE technology, while some users can support Ma-MIMO technology. The number of active users who can only be served by an LTE AP or Ma-MIMO AP is denoted by K^L and K^{Ma} respectively. The overlapped area between Ma-MIMO and LTE cells has K^{M-RAT} users who can be assigned to either Ma-MIMO or LTE APs. The overall number of active users who can be served by Ma-MIMO AP is denoted by K .

The massive-MIMO base station is equipped with a large number of antennas (M) serving a set of active users K . The system operates in TDD mode, where each UE transmits a frequency-orthogonal uplink (UL) pilot for the channel estimation. The estimated channel matrix between the UEs and the BS is denoted by $\hat{\mathbf{H}} \in \mathbb{C}^{M \times K}$. While the actual channel matrix during the uplink data transmission is denoted by $\mathbf{H} \in \mathbb{C}^{M \times K}$, which is given by

$$\mathbf{H} = \hat{\mathbf{H}} + \mathbf{E} \quad (1)$$

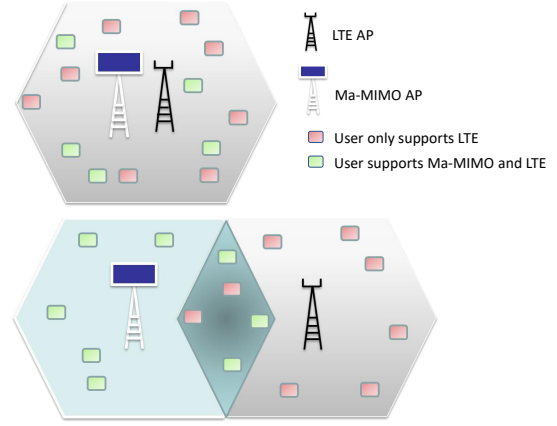


Fig. 1. Massive MIMO and LTE cell architecture.

where $\mathbf{E} \in \mathbb{C}^{M \times K}$ is the difference between the estimated channel and the actual channel during the uplink data transmission. This error could be caused by hardware impairments, interpolation across frequency, large-scale channel attenuation and any other potential sources. In this paper the error in the channel estimation between the UE and the BS is modeled as a complex Gaussian distribution $\sim \mathcal{CN}(\mathbf{0}, \sigma_e^2 \mathbf{I}_M)$, where $\mathbf{I}_M$ is the $M \times M$ identity matrix and σ_e^2 is the error variance. Whilst this model may not be so accurate in reality, it serves to illustrate the potential effects of channel state information (CSI) inaccuracies in Ma-MIMO. The equalized UL signal $\hat{\mathbf{x}} \in \mathbb{C}^K$ can be expressed as

$$\hat{\mathbf{x}} = \mathbf{W}(\sqrt{\rho_{ul}} \mathbf{H} \mathbf{x} + \mathbf{n}) \quad (2)$$

where $\mathbf{x}$ represents the transmitted symbol vector from all users in the same cell, normalized as $\mathbb{E}\{|x_k|\} = 1$. The corresponding UL transmit power is denoted by ρ_{ul} . Simple UL power control is assumed in this paper. The UL transmit power is adjusted so the received signal to noise ratio (SNR) from all users is the same. The additive noise vector is denoted by $\mathbf{n}$. The noise variance from the antennas at the BS is modeled as $\sim \mathcal{CN}(\mathbf{0}, \sigma_n^2 \mathbf{I}_M)$, where σ_n^2 is the noise variance. $\mathbf{W} \in \mathbb{C}^{K \times M}$ is the linear decoder matrix, formed using Minimum Mean Square Error (MMSE), zero-forcing (ZF) or matched filtering (MF).

III. SYSTEM OVERVIEW

This section describes a high-level overview of an SDN-RRM framework, one of the key contributions of this paper. Fig 2 provides a high-level schematic of the proposed framework, which consists of four main components: RRM applications, a control plane, SDN agents and data planes. In the application plane, each RRM module represents an RRM algorithm implementation, which makes resource allocation decisions based on the performance metrics obtained from the agent and QoS requirements of different traffics. The SDN controller

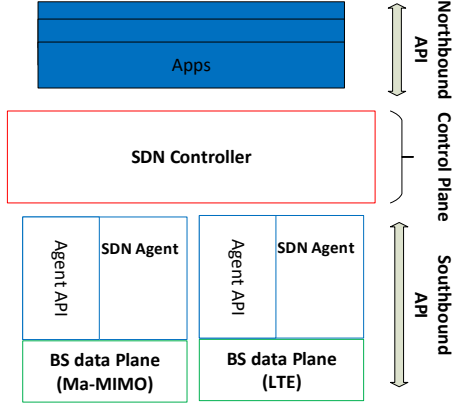


Fig. 2. High-level schematic of SDN-RRM framework.

is the main component of the control plane, linking the RRM applications and the SDN agents. In terms of RRM, the controller is in charge of delivering the RRM decisions from RRM applications to SDN agents and providing RRM applications with an abstract view of the network, for example through statistics and events. The control plane and data plane separation is enabled by the SDN agent and its API.

The main functionality of the SDN agent is to facilitate communications between the SDN controller and wireless network element (NE) by establishing a two-way channel between them. In one direction, the agent receives and processes RRM instructions from the SDN controller and passes them to the data plane, while in the other direction the agent constructs and sends response to the SDN controller along with statistics of the underlying wireless networks.

The Wireless NE is a logical entity in the data plane, representing the Medium Access Control (MAC) and physical layer (PHY) functionalities of a particular wireless access technology. One of the main function modules in NE is the forwarding/traffic processing engines that decide how the traffic flows are to be handled. With regard to RRM, this is where the SDN agent applies the RRM decision made by RRM application to the base station. In other words, the RRM application affects the behaviour of the NE by using the SDN agent. Each network element has its own SDN agent, while the SDN controller connects multiple SDN agents.

IV. MULTI-RAT SWITCHING METHODOLOGY

The switching decision between Ma-MIMO and LTE APs aims to maximize the throughput (R) of the network and achieves the minimum throughput (R^{min}) requirements for all users. The key is to mitigate the interference from the estimated channel coefficients between users and also the interference caused by inaccurate CSI in

Ma-MIMO. This is achieved by relying on the Error Vector Magnitude (EVM) prediction technique in [10] and [11]. The EVM is a common measurement of signal quality used in 3rd Generation Partnership Project (3GPP) LTE standards [4]. This is a comprehensive metric because it considers all impairments upon the transmitted symbol as seen at the receiver. Higher modulation and coding schemes (MCS) are supported when the EVM is small. The EVM can be found as [12]

$$EVM_{RMS} = \sqrt{\frac{\frac{1}{N^s} \sum_{n=1}^{N^s} |S_r(n) - S_t(n)|^2}{\frac{1}{N^s} \sum_{n=1}^{N^s} |S_t(n)|^2}} \quad (3)$$

where N^s is the number of symbols the EVM was measured over. $S_r(n)$ is the n th normalized received symbol and $S_t(n)$ is the ideal value of the n^{th} symbol.

The multi-RAT switching methodology relies on the CSI, SNR and EVM performance metrics from Ma-MIMO and LTE APs. The SDN controller uses these metrics to decide the scheduling and the wireless AP for each user by following the flowchart in Fig 3. The algorithm starts by serving all Ma-MIMO users in $\mathcal{U}^{ma}$ simultaneously in one group ($N = 1$) using the same frequency. These users belong to the Ma-MIMO cell and are not overlapped with the LTE cell. If the QoS for any user is below the minimum requirement, additional Ma-MIMO group will be created. Users in different Ma-MIMO groups are served either in different time slots or different frequencies. This stage is fully described in Algorithm 1, which is covered later in this section. Users who can not support Ma-MIMO are assigned to the LTE AP. These users belong to the LTE cell, where some of them are overlapped with the Ma-MIMO cell. The rest of the users are overlapped between the LTE and the Ma-MIMO cells and have the capability to support both technologies (users in $\mathcal{U}^{M-RAT}$). The SDN controller assigns the users in $\mathcal{U}^{M-RAT}$ either to one of the Ma-MIMO groups or to the LTE AP. The assigning process is described in Algorithm 2, which is covered later in this section. If R is below R^{min} for any user, additional Ma-MIMO group will be created and go to Algorithm 1. When all users achieve the minimum throughput requirement, some Ma-MIMO users can be assigned to more than one group to maximize the throughput (condition to $\forall R_a^{all} \geq R^{min}$).

The predicted EVM ($\widetilde{EVM}$) technique introduced in [11] and [10] is adopted by the SDN controller. This method jointly relies on the channel correlation between users and the EVM performance to predict the EVM values for a subset of the active users without the need to retransmit

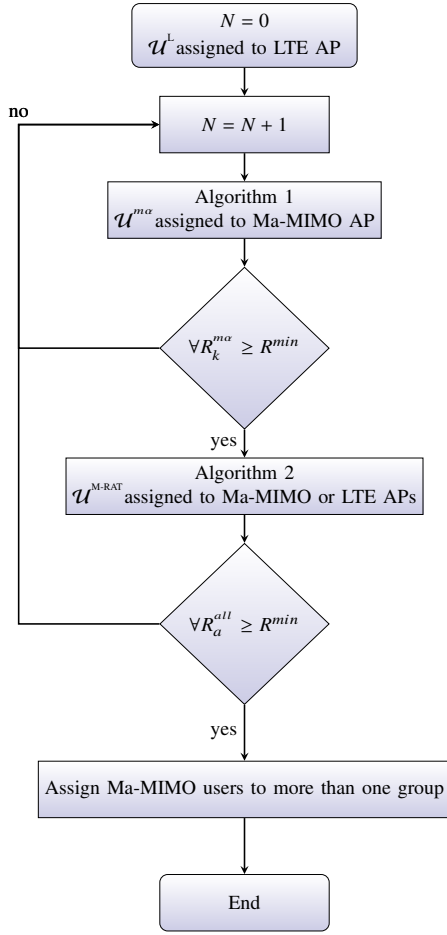


Fig. 3. Flowchart of the multi-RAT switching algorithm.

data symbols. This method considers the user channel vector and the interference caused by inaccurate CSI. In addition to the $\widetilde{\text{EVM}}$, the SDN controller relies on the following definitions:

Definition IV.1. $v(\mathcal{A}, k)$ is the average $\widetilde{\text{EVM}}$ for the total number of users $\tilde{K}$, where:

- $\mathcal{A}$ is a group of users.
- k is the candidate user to join the group $\mathcal{A}$.
- $\tilde{K}$ is the total number of users in the set $(\mathcal{A} \cup \{k\})$.
- $v(\mathcal{A}, k) = \frac{1}{\tilde{K}} \sum_{i=1}^{\tilde{K}} \widetilde{\text{EVM}}_i$.

Definition IV.2. The predicting R ($\tilde{R}$) is the estimated R for a set of users calculated by using the highest MCS mode the $\widetilde{\text{EVM}}$ allows. This calculations are under an error free assumption considering 3/4 rate low-density parity-check (LDPC) [13].

Definition IV.3. $\psi(\mathcal{A}, k)$ is the $\tilde{R}$ for a set of users in group $\mathcal{A}$ and the candidate user to join the group (k), where the $\widetilde{\text{EVM}}$ for each user is calculated based on new set of users $(\mathcal{A} \cup \{k\})$.

A. Algorithm 1: Assigning process for users in $\mathcal{U}^{ma}$

Users in $\mathcal{U}^{ma}$ are assigned into one of the Ma-MIMO groups $\mathcal{G}_n$ as described in Algorithm 1, where $n \in \{1, \dots, N\}$. Users with the highest spatial correlation are first assigned into different groups. Then the rest of the users are assigned to one of the groups based on maximizing the $\tilde{R}$, minimizing the average $\widetilde{\text{EVM}}$ and achieving the minimum throughput requirement. By minimising the average EVM values in each group, interference caused by the user channel vectors and the hardware impairments is decreased. This means higher throughput can be achieved. If the throughput for any user is below R^{min} , additional group will be created as it is shown in the flowchart in Fig 3. If increasing number of groups couldn't achieve the minimum throughput requirement, users with highest EVM values will be removed and won't be considered in the algorithm.

B. Algorithm 2: Assigning process for users in $\mathcal{U}^{M-RAT}$

The remaining users are assigned to one of the Ma-MIMO groups or to the LTE AP, as described in Algorithm 2. $\mathcal{G}^{M-RAT}$ is the group vector for Ma-MIMO and LTE APs, where $\mathcal{G}^{M-RAT} = \mathcal{G}^{ma} \cup \mathcal{G}^L = \{\mathcal{G}_1^{ma}, \dots, \mathcal{G}_N^{ma}, \mathcal{G}_{N+1}^L\}$. Since only one LTE AP is considered in this paper, $\mathcal{G}^L$ is only one group. Users in $\mathcal{U}^{M-RAT}$ are assigned to $\mathcal{G}_r^{M-RAT}$, where $r \in \{1, \dots, N, N+1\}$. This means that users are assigned to one of the Ma-MIMO groups (group index from 1 to N) or to the LTE AP (group index = $N+1$). Users are assigned to one of the groups based on maximizing the $\tilde{R}$, minimizing the average $\widetilde{\text{EVM}}$ and achieving the minimum throughput requirement. If the throughput for any user is below R^{min} , additional Ma-MIMO group will be created as it is shown in the flowchart in Fig 3. If increasing number of groups couldn't achieve the minimum throughput requirement, users with high EVM values will be removed and won't be considered in the algorithm.

V. RESULTS USING WIRELESS SDN SIMULATOR

In this section, the proposed SDN controller framework is evaluated using the wireless SDN simulator developed at the University of Bristol. For comparison purposes, the conventional handover concept between 5G and fourth generation (4G) or 4G and third generation (3G) is applied using the Ma-MIMO simulator and the LTE simulator developed by the University of Bristol. Generally, the user decides which wireless AP technology to use in this method. The decision is taken based on the received signal strength at the user side. When the radio resources are not enough, the BS tells the user to switch to another wireless AP [14]. independent and identically distributed (IID) Rayleigh channels were used in both scenarios, where ZF technique is used as an equalizer for Ma-MIMO and LTE. MU-MIMO is used with four antennas

Algorithm 1: $\mathcal{U}^{ma}$ assigned to Ma-MIMO AP

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1 set  $\mathcal{G} = \{\}, \mathbf{G} = \left| \hat{\mathbf{H}}^{maH} \hat{\mathbf{H}}^{ma} \right|, T = \{1, \dots, N\}$ .
2 %Assigning one user for each group.
3 for  $n = 1 : \lceil N/2 \rceil$  do
4     Find the highest correlation value:  $g_{i,j}$ 
5     if  $(n = \lceil N/2 \rceil \neq N/2)$  then
6         ▶ This condition assigns one user to
           group number  $N$ , when  $N$  is an odd
           number.  $l \leftarrow \operatorname{argmin}_{l \in \{i,j\}} \widetilde{EVM}_l$ 
7          $\mathcal{G}_N = \mathcal{G} \cup \{\mathcal{U}_l^{ma}\}$ 
8         Remove  $\mathcal{U}_l^{ma}$  from  $\mathcal{U}^{ma}$ 
9     else
10        ▶ Assigning users with the highest
          correlation to different groups.
11         $\mathcal{G}_{(n \times 2) - 1} = \mathcal{G}_{(n \times 2) - 1} \cup \{\mathcal{U}_i^{ma}\}$ 
12         $\mathcal{G}_{(n \times 2)} = \mathcal{G}_{(n \times 2)} \cup \{\mathcal{U}_j^{ma}\}$ 
13         $g_{i,j} = 0$ , Remove  $\mathcal{U}_i^{ma}$  &  $\mathcal{U}_j^{ma}$  from  $\mathcal{U}^{ma}$ 
14    end
15 %Assigning the remaining users to one of the
    groups
16 for  $k = 1 : (K^{ma} - N)$  do
17      $l \leftarrow \operatorname{argmax}_{l \in T} \psi(\mathcal{G}_l, \mathcal{U}_k^{ma}) : \forall R \geq R^{min}$ 
18      $p \leftarrow \operatorname{argmin}_{p \in l} \nu(\mathcal{G}_p, \mathcal{U}_k^{ma})$ 
19      $\mathcal{G}_p = \mathcal{G}_p \cup \{\mathcal{U}_k^{ma}\}$ 
20     ▶ User is assigned to the group based on
       achieving the max  $\widetilde{R}$ . If assigning the user
       to more than one group achieves the same
       max  $\widetilde{R}$ , minimising the average  $\widetilde{EVM}$  of the
       group is considered.
21     Remove  $\mathcal{U}_k^{ma}$  from  $\mathcal{U}^{ma}$ 
22 end
    
```

at the LTE AP. While 128 antennas are used at the Ma-MIMO AP. The total number of active users is a multiple of $8 \in [16, 88]$. At first, the total number of active users is 16. This is kept on increasing by 8 users until it reaches 88 active users. Half of the total active users is located at the Ma-MIMO cell without overlapping with the LTE cell. The second half is overlapped between the Ma-MIMO and the LTE cells. All users are capable of supporting Ma-MIMO and LTE technologies. 3/4 rate LDPC code was used in the simulator. A throughput of 8 Mbps is set as a QoS requirements. This achieves the recommended speed from Microsoft to use Skype for video calling with High-Definition (HD) quality or for a group video teleconference [15].

Algorithm 2: $\mathcal{U}^{M-RAT}$ assigned to Ma-MIMO or LTE APs

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1  $R_a^{all}$  is  $\widetilde{R}$  from Ma-MIMO and LTE APs, where  $a \in$ 
    $\{1, \dots, K^{M-RAT} + K^{ma} + K^L\}$ .
2 set  $\mathcal{U}^{tmp} = \mathcal{U}^{M-RAT}, T = \{1, \dots, N + 1\}$ 
3 for  $r = 1 : K^{M-RAT}$  do
4      $l \leftarrow \operatorname{argmax}_{l \in T} \psi(\mathcal{G}_l^{M-RAT}, \mathcal{U}_r^{M-RAT})$ 
        $:\forall R_a^{all} \geq R^{min}$ 
5      $p \leftarrow \operatorname{argmin}_{p \in l} \nu(\mathcal{G}_p^{M-RAT}, \mathcal{U}_r^{M-RAT})$ 
6      $\mathcal{G}_p^{M-RAT} = \mathcal{G}_p^{M-RAT} \cup \{\mathcal{U}_r^{M-RAT}\}$ 
7     ▶ The Ma-MIMO group or the LTE AP is
       selected based on achieving the max  $\widetilde{R}$ . If
       assigning the user to more than one group
       achieves the same max  $\widetilde{R}$ , minimising the
       average  $\widetilde{EVM}$  of the group is considered.
8     Remove  $\mathcal{U}_r^{M-RAT}$  from  $\mathcal{U}^{M-RAT}$ 
9 end
    
```

Fig. 4 shows the total throughput from all active users achieved by using Ma-MIMO and LTE APs. When total number of users is 16, the proposed wireless SDN approach increased the throughput by 37 Mbps. When increasing the number of active users, the throughput gains by using the SDN approach kept increasing compared to the conventional handover approach. This is because users with high spatial correlation are either assigned to different Ma-MIMO groups or to the LTE AP. While in the conventional handover approach, users always start by joining the Ma-MIMO AP. The LTE AP is only used when the Ma-MIMO received signal strength at the user side is poor or when the Ma-MIMO radio resources are not enough to serve the available users. When total number of active users is 88, the RRM application noticed that the current radio resource can not serve all users. Therefore, eight users with the highest EVM value were removed in order to achieve the QoS requirement, and the total throughput is increased by 477 Mbps compared to the conventional handover approach. Besides, the minimum throughput achieved by any user was always above the QoS requirements, as it is shown in Fig. 5. When the conventional handover approach is used, some users could not achieve the QoS requirements. This can be seen by increasing the total number of active users above 28. Furthermore, users with zero throughput start appearing when total number of active users is increased above 50. When all active users are served, 27 user couldn't achieve the QoS requirements, as it is shown in Fig. 6.

VI. REAL-TIME VALIDATION USING TEST-BEDS

In this section, the proposed SDN framework is evaluated in real-time by using the massive MIMO testbed

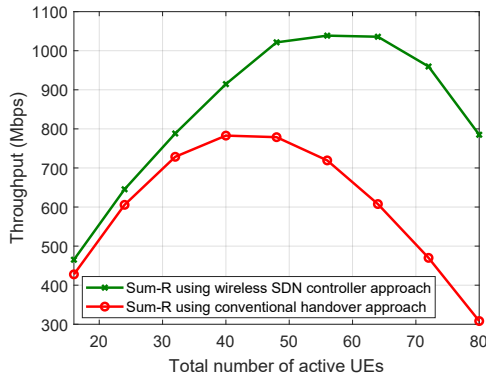


Fig. 4. Total throughput from all active users.

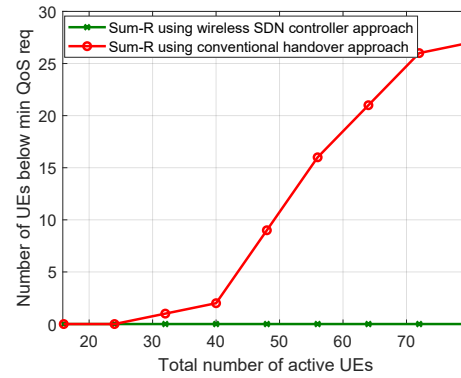


Fig. 6. Number of users who couldn't achieve the QoS requirements.

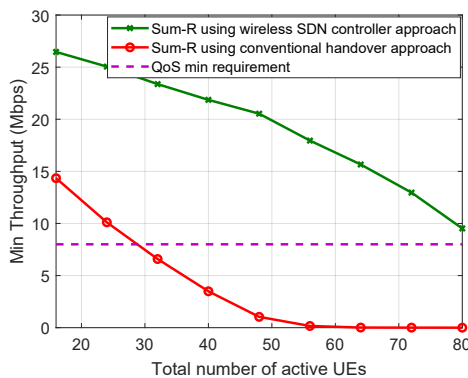


Fig. 5. Minimum throughput achieved by any active user.

in [3], the LTE testbed in [16] and the SDN controller testbed. The whole experiment was recorded in [17], which can be seen on the second half of the video. We use Ryu [18] SDN controller as the control plane in our framework and build our RRM applications using the northbound API provided by Ryu. For practical reasons, we implement our SDN agents in the data plane on the National Instruments (NI) universal software radio peripheral (USRP) platform. The agent is developed in LabView and sits right on top of higher MAC in LTE and Ma-MIMO, which means that the added agent functions will not affect the performance of lower-layer Tx/Rx tasks. The physical connection from the agent to the controller is via Gigabit Ethernet cables.

The Communication Systems & Networks (CSN) lab was used for line-of-sight (LOS) measurements between the testbeds and 12 UEs from 6 USRPs. An overview of the setup can be seen in Fig. 7. At the Ma-MIMO BS side, 128 element array was used providing half-wavelength spacing at 3.5 GHz and ZF equaliser is used. The conventional handover approach is also evaluated in this experiment. A floor plan of the experiment is shown in Fig. 8 with the wireless APs, the SDN controller and the UEs locations. The user identity (ID) of the first UE

on the left in Fig. 8 is one. By moving to the right, the user ID is increased by one for each UE. Attenuators were added to all users in order to reduce the transmitted and received signal strength. The effect of this would be almost equivalent to increasing the distance between users and the wireless APs. A throughput of 20 Mbps is set as a QoS requirement. This achieves the minimum requirement for 4K quality video [19].

Table I shows the real-time throughput for both scenarios. When the conventional handover approach was used, a total throughput of 134.2 was achieved. This value was increased when the wireless SDN controller approach was used by 48.4 Mbps. Furthermore, the minimum throughput achieved by any user was 20 Mbps. This value was decreased by 5.9 Mbps when the conventional handover approach was used, which is below the minimum QoS requirements. The wireless SDN controller approach outperforms the conventional handover approach due to the radio resource allocation between users. Since the conventional handover approach mainly relies on the received signal strength at the user side, the eight users next to the Ma-MIMO BS were using Ma-MIMO as a wireless AP. The four users next to the LTE BS were using the LTE as a wireless AP. When the wireless SDN approach was used, the spatial correlation between users was carefully considered. Users 3,4,7,8,9,10,11 and 12 were assigned to the Ma-MIMO AP and users 1,2,4 and 6 were assigned to the LTE AP. Although users 3,4,9,10 were closer to the Ma-MIMO AP, they were assigned to the LTE AP in order to reduce the interference between adjacent users due to the high spatial correlation level.

VII. CONCLUSIONS

In this paper, we proposed a wireless SDN framework for radio resource allocation between a single cell LTE and a single cell Ma-MIMO. The framework has a flexible and programmable platform for the implementation of RRM algorithms in a multi-RAT environment. The proposed

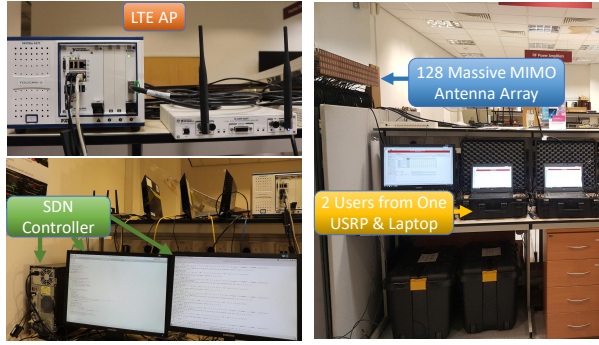


Fig. 7. Main hardware components of the spatial SDN controller framework.

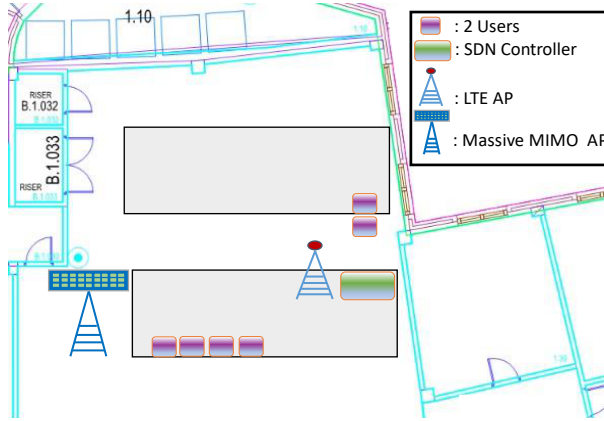


Fig. 8. CSN Lab floor plan showing the wireless APs, SDN Controller and users locations.

TABLE I
REAL-TIME THROUGHPUT.

	Total R (Mbps)	min R per UE (Mbps)
Wireless SDN controller	182.6	20
Conventional handover	134.2	14.1

approach enables the separation of the control plane from the data plane through a customized agent and its APIs, while providing real-time RRM operations in multi-RAT networks. The SDN controller decides the number of Ma-MIMO groups, where users are served in different time slots or frequencies. Then it assigns users to either one of the Ma-MIMO groups or to the LTE AP in order to maximise the capacity and achieves the QoS requirements. The proposed SDN framework was built and evaluated. Real-time comparison was conducted between the proposed framework and the current conventional handover concept, where the user mainly decides which wireless AP to use. The wireless SDN approach outperformed the conventional handover approach by delivering higher capacity gains. Furthermore,

the QoS requirement was achieved for all users, unlike the conventional handover approach.

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